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Biomedical image-based keratoconus classification using convolutional transformers and grey goose optimization

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ABSTRACT

Background: Keratoconus (KCN) is a progressive corneal disorder that leads to visual distortion and increases the risk of complications during refractive surgical procedures. Accurate and early diagnosis is critical for effective clinical management.
Objective: This work presents a novel deep learning model for keratoconus classification from corneal topographic images.
Problem: Conventional diagnostic models are based on corneal topography and biomechanical properties. These processes are time-consuming and prone to variability.
Methodology: This work presents an optimization based Deep Learning (DL) model Grey Goose Optimization (GGO) based Convolutional Transformer (CT) model for KCN classification. The model considers the feature extraction capabilities of convolutional neural network (CNN) and the attention mechanism of transformers for enhancing classification accuracy. The GGO is integrated for fine-tuning the hyperparameters of the model and improving its robustness.
Findings: The proposed approach is evaluated on a Keratoconus detection dataset and shown superior performance compared to conventional DL models. The outcomes shows the robustness of the suggested approach in correctly categorizing KCN by achieving accuracy of 99.2% and precision of 98.9%
Conclusion: The proposed CT-GGO provides the way for automatic and reliable diagnostic tools in ophthalmology.
Recommendations: Future research will demonstrate cross-center validation, larger data, and multi-modal inputs for improving model generalization and interpretability.

1. Introduction

Keratoconus (KCN) is an increasing corneal condition characterized by thinning of the cornea and it causes impaired vision, protrusion, and abnormal astigmatism. The development of KCN is impacted due to the genetic condition and environmental triggers [1]. General risk terms associated with the condition like eye rubbing, atopy, and prolonged ultraviolet radiation exposure. In addition, studies have reported familial clustering of KCN, and showing a potential hereditary component. During the last decades, emergence has been made in both the diagnosis and treatment of KCN [2]. While it was once considered a rare disorder, it was previously estimated by the National Institute of Health to affect approximately 1 in 2,000 people. In the context of corneal refractive surgery, undiagnosed KCN is considered major risk criteria for iatrogenic keratectasia and it is one of the extreme and irreplaceable complexities following the surgeries [3]. As a result, precise KCN identification of KCN is essential during pre-surgery screening for ensuring patient safety.

Misdiagnosis and delayed identification of KCN, in severe cases, lead to vision defect. Identifying KCN in a suspected eye is challenging and necessitates a comprehensive assessment before performing refractive surgery [4]. Hence, proper diagnosis is important for effective case management and timely surgical intervention when required. This approach prevents further vision worsening and ensures that any refractive errors remain manageable. This allows for correction with simple solutions like vision lenses after surgery. Proper detection of KCN is essential in managing its progress and timely diagnosis allows for the treatment process [5].

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Recent times, Machine Learning (ML) model are better at gaining both the accuracy and efficiency of KCN diagnosis and predicting progression. Conventional diagnostic approaches are based on expert evaluation of corneal imaging and biomechanical properties [6]. It sometimes leads to subjective interpretation and delayed diagnosis. ML models like Support Vector Machines (SVM), Random Forests (RF), and Convolutional Neural Networks (CNNs) are generally utilized ML models [7]. SVM can categorize KCN by utilizing handcrafted features. SVM models can differentiate between normal and KCN by identifying complex patterns in curvature and elevation maps [8]. RF has been utilized for improving diagnostic robustness by integrating different decision trees trained on various subsets of corneal imaging features. This method enhances classification accuracy and minimizes the overfitting risks [9].

The deep learning (DL) based approaches like CNN have transformed KNC detection by automatically learning complex patterns from high-dimensional corneal images without manual feature extraction [10]. DL models can assist ophthalmologists by increasing diagnostic accuracy and recommending treatment plans. Further, it helps in reducing the time needed to analyze the images of affected eyes and allows for more efficient patient care. CNN models can extract hierarchical representations of corneal structures and detect even fine abnormalities indicative of early KNC [11 25]. These advancements in improved early diagnosis and contributed to predicting disease progression [12]. Models like ResNet, EfficientNet and Vision Transformers (ViTs) [13] can be trained on labeled data for classifying normal and KCN cases. Further, Transfer learning with pre-trained models generally improves accuracy because of the limited medical data [14]. The CNN models are effective at extracting localized spatial features and they generally fall short in modeling long-range dependencies. This is important to understand complex deformations in the corneal shape. Then, ViTs needs large data and lack inductive biases. This makes them less ideal for medical applications with limited data. The Convolutional Transformer (CT) model combines the advantages of architectures by incorporating convolutional layers for local feature extraction and transformer blocks for global contextual features. This enhanced model ensures better representation learning and is important for accurate KCN detection. The major contributions are:

- To introduce Convolutional Transformer (CT) model optimized with GGO, combining CNNs for feature extraction and transformers for enhanced attention based classification.
- To present the integration of GGO for fine-tuning hyperparameters and this enhances model robustness and classification accuracy.
- To evaluate on Keratoconus detection dataset and shows better accuracy compared to other model and ensure an effective diagnosis.
- To perform a comprehensive ablation study across multiple training splits for validating the stability of the CT-GGO model under different data availability conditions.

Research Questions (RQs).

This study overcomes the following research questions:

RQ1: Can hybrid Convolutional Transformer (CT) model improve the classification accuracy of keratoconus (KCN) effectively from corneal topographic images compared to other models?

RQ2: How does hyperparameter tuning using Grey Goose Optimization (GGO) influence the performance and generalization of the proposed CT model?

RQ3: Can the CT-GGO model distinguish between Normal, Suspect, and KCN cases in imbalanced real-world clinical datasets?

RQ4: Can the interpretability of the model align with clinical expectations to support decision-making in real-world ophthalmic settings?

The rest sections are: Section 2 shows the existing works based on the KCN detection; Section 3 shows the proposed KCN detection with diagrammatic and mathematical explanation; Section 4 illustrates the outcomes and Section 5 ends the work.

2. Related works

A DL approach to identify the base curve (R0) of rigid gas permeable contact lenses for KCNs patients [6]. Using data from 358 eyes, a U-net architecture was trained on anterior axial curvature maps from Scheimpflug imaging. The model attained a higher R^2 value (0.83) and lower mean absolute error (0.16 ± 0.13). Multivariate analysis showed that an off-center apex increased prediction error and this existing approach offered enhanced accuracy and could simplify complex lens fitting. A novel method for diagnosing and treating various kind of KCN by applying few shot learning (F-SL) combined with DL models by the Keraring nomogram and corneal topographic images. Conducted as a retrospective cross-sectional analysis, it involved 268 corneal images from 175 patients who received Keraring segment implants between May 2020 and September 2022. The research introduced several pre-trained models and a prototypical model for detecting and classifying corneal abnormalities[15].

The researchers [16] considered dataset comprises images from hospitals, SynthEyes, and GANs. Pre-processing undergoes resizing and normalization. The data was split into training, validation, and test sets. Different DL models Custom CNN, CNN-SVM, CNN-LSTM, RESNET50, Xception, MobileNetV2, and VGG19 were considered. Validation ensured optimized hyperparameters and overcomes overfitting. A Multi-disease DL model for corneal diseases detection. Here, the 1, 58,220 anterior segment optical coherence tomography (ASOCT) images were considered and the data was collected between the years 2016 to 2019. Then, the pre-trained VGG 19 was classifying corneal disease and normal. Then, the model was tested in the additional independent setting [17]. A comparison between ML and a DL approach called CorNet for processing Corvis ST raw data [18]. The ML pipeline has handcrafted feature extraction, followed by prior feature selection and diverse classification and lead to final classification outputs such as CBI and cCBI. Then, CorNet automates feature extraction using a CNN and the linear softmax regression layer for classification. The CorNet streamlines the process and enhances performance by learning features directly from raw data. The author [19] designed convolutional block was to extract diverse image features and maintained spatial information. Then, a DL model was developed by enhancing the existing BiO-Net architecture to enable the simultaneous segmentation of three distinct corneal micro-layers. The data has 1, 712 OCT images was and all of them were annotated for identifying the 3 micro-layers for segmentation tasks. [20] focused on detecting the three primary corneal layer boundaries by the capsule network. A hybrid speckle noise reduction filter was used for reducing noise in corneal images. Then, the ClassCaps algorithm was exploited for training and selecting the best noiseless image from a set of test images. Finally, the modified SegCaps was employed for identifying the three main corneal boundaries from the denoised image.

A CNN model for KCN detection by corneal deformation videos. A dataset has 734 non-consecutive individuals and has the data of refractive surgery patient, unilateral KNC and bilateral KNC patients [21]. A video preprocessing pipeline was developed for converting deformation videos into 3D pseudoimage representations. The probability score thresholding was determined, and its performance was assessed.

An ensemble of Deep Transfer Learning (EDTL) model exploiting corneal topographic maps. Four pre-trained models like SqueezeNet, AlexNet, ShuffleNet, and MobileNet-v2 were considered [22]. Then, the Pentacam indices (PI) were integrated and the EDTL fused the probability outputs from these classifiers for enhancing decision-making. The PI classifier attained an accuracy of 93.1 % and the individual deep DL model attained 90 % accuracy only in certain instances. Seven DL models [24] was introduced and each model was trained on different corneal map types like anterior/posterior eccentricity, elevation, sagittal curvature, and corneal thickness for capturing rich structural features.

The existing works are based on handcrafted features and standard CNN models, which may fail to generalize to diverse or imbalanced datasets. Others need large labeled data and use complex ensemble models that are computationally expensive and complex to interpret. However, the suggested model overcomes these limitations by considering a unified and end-to-end DL model that combines the spatial inductive bias of CNNs with the contextual modeling capabilities of transformers. The integration of GGO further differentiates the suggested model which offers a robust and automated hyperparameter tuning mechanism not previously demonstrated in this domain. These differences clearly show the novelty and contribution of the suggested research.

3. Proposed methodology

KCN is an increasing corneal disorder that leads to thinning and bulging of the cornea and causes visual impairment. Proper KCN detection is important for better disease management. This work presents a CT- GGO model for the classification of KCN. The model integrates the feature extraction abilities of CNN with the attention mechanism of transformers are presented for classification enhancement. GGO is presented hyperparameter tuning of the suggested DL model. Fig. 1 shows the proposed structure of the KCN detection model. The pipeline is composed of five key stages: image preprocessing, augmentation, feature extraction using convolutional layers, hierarchical encoding via transformer blocks, hyperparameter optimization via GGO and final classification.

3.1. Pre-processing

The image resizing is important in KCN disease detection and ensures an efficient DL performance. Generally, corneal topographic maps have different resolutions, which can lead to inconsistencies in feature extraction. In this work, resizing standardizes these images to a fixed dimension of 256 × 256 pixels and ensures uniform input to CT based models. The size of 256 × 256 pixels is chosen after empirical experimentation with multiple resolutions like 128 × 128, 224 × 224, and 512 × 512. The chosen size offered an optimal trade-off in preserving important structural details and optimizing computational efficiency. This resolution ensures that essential topographic features are retained and provides better training of DL models with reduced memory and processing requirements.

3.2. Augmentation

The stratified sampling is employed during train-test splits for preserving class proportions across each evaluation stage. Here, the data augmentation approaches like random rotations, horizontal and vertical flips, intensity scaling, and slight geometric transformations are performed. These transformations preserve important structural features and improve model generalization. The, the class-weighted loss function is applied during training for penalizing misclassification of minority classes more heavily and ensures balanced learning. These models improve the ability of the CT-GGO to correctly find all three classes and minimize bias toward dominant categories. Further, it supports more accurate and equitable KCN classification.

3.3. Feature extraction and classification

Fig. 2 shows the structure of the CT model and it is used for extracting and classifying the Normal, KCN and suspect cases. In the Feature extraction Phases, where the input image undergoes progressive transformations for extracting essential features for classification. The process initiates with an input image and is provided through a sequence of four distinct phases and it has downsampling and CT blocks. These phases gradually minimize the spatial image dimensions and increasing its depth. This makes the model for capturing both low-level and high-level features. In Phase 1, the input is downsampled by a factor of 4,

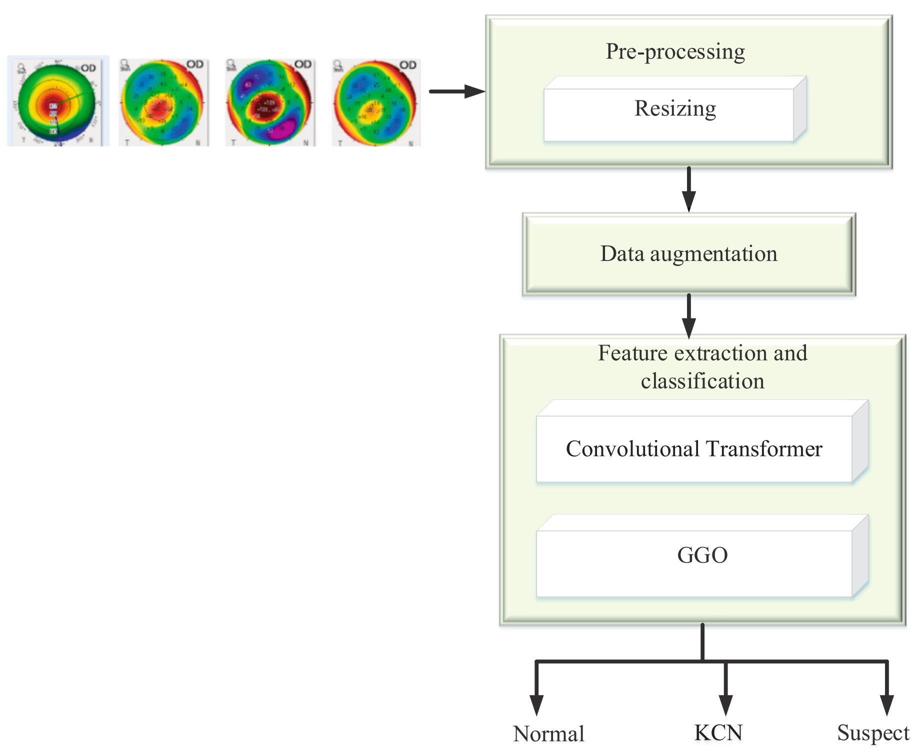

Fig. 1. Proposed structure of the KCN detection model.

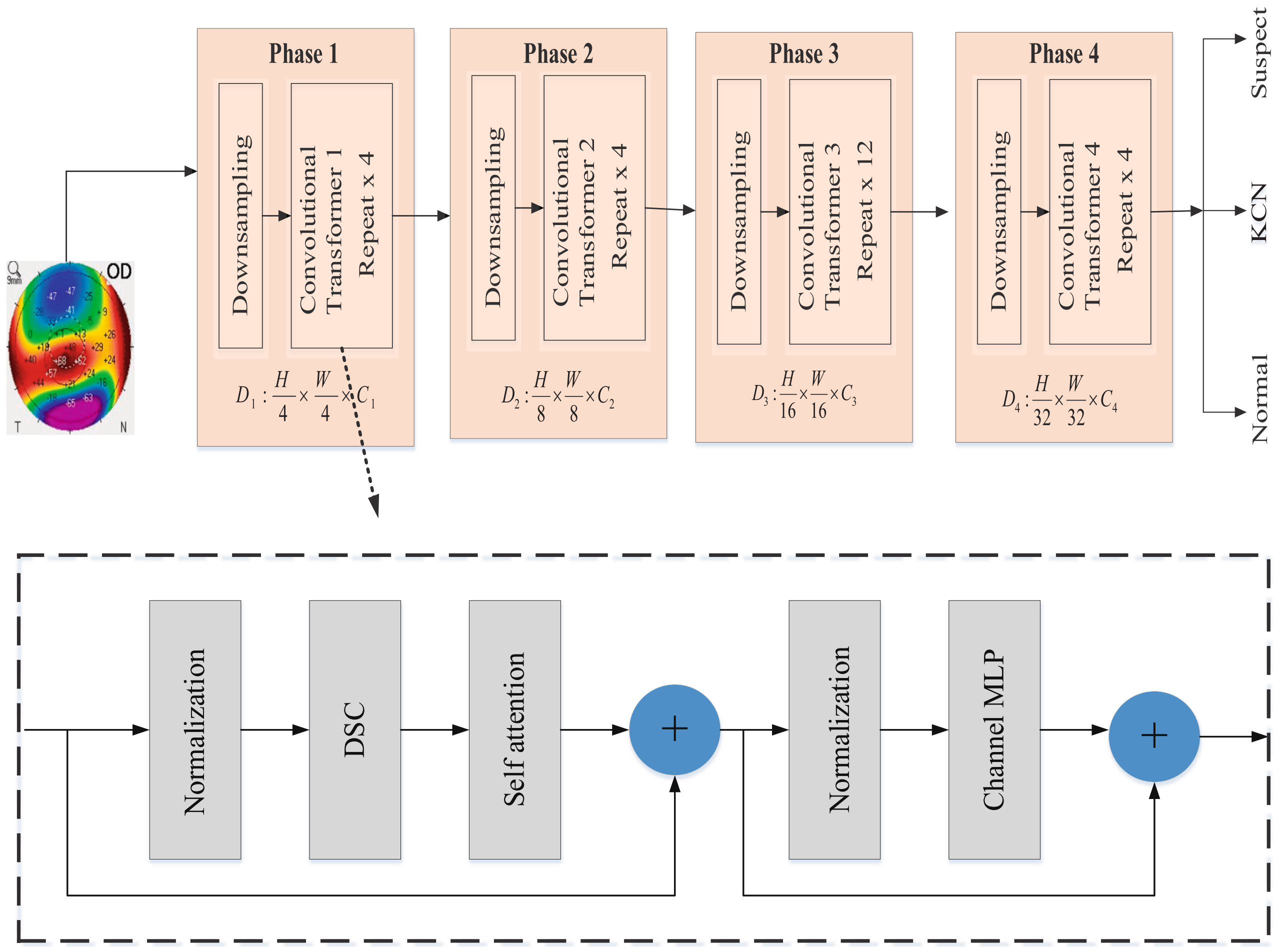

Fig. 2. Structure of the CT model.

and a CT block is repeated four times to extract initial feature representations. Moving to Phase 2, the resolution is further reduced to $\frac{H}{8} \times \frac{H}{8}$ and additional transformer blocks refine the extracted features. In the Phase 3, the resolution is decreased to $\frac{H}{16} \times \frac{H}{16}$ and repeating the transformer block 12 times. This allows the network for capturing more complex patterns. Finally, in Phase 4, the resolution is reduced to $\frac{H}{32} \times \frac{H}{32}$. This ensures that only the most essential features remain. Every phase increases feature representations, reduces redundancy and making the extracted information more suitable for classification. The output from the final phase is exploited to determine the three categories: Normal, KCN (likely Keratoconus), or Suspect.

The suggested CT model has the phases such as patch embedding, depth wise separable convolutions (DSC), attention mechanism and classifier head. The input image y is passed through a convolutional layer that utilizes a kernel size and stride of 4x4. It is expressed as:

$$patch\_embedding = Convolution\_2D(y, kernel\_size = 4, stride = 4) \quad (1)$$

All four phases utilize DSC to process the feature maps and it is expressed as:

$$DSC(y) = ReLU(Convolution\_2D)(y, kernel\_size = 3, C, padding = 1) \quad (2)$$

$$Point\_wise(y) = ReLU(Convolution\_2D)(y, kernel_{size} = 1) \quad (3)$$

where C is the total channels. After the DSC, the attention mechanism is incorporated for capturing long range dependencies in the feature maps. The attention mechanism is given as:

$$At(Q_e, K_e, V_a) = Softmax\left(\frac{Q_e K_e^T}{\sqrt{d_k}}\right) V_a \quad (4)$$

where Q_e, K_e, V_a is the Query, Key, Value; d_k is the key vector's dimensionality. At last, the classification is performed which has Global Average Pooling (GAP), a Fully Connected (FC) layer and SoftMax. It is expressed as:

$$Output(y) = Softmax(FC(GAVP(y))) \quad (5)$$

The GGO [23] algorithm optimizes the hyper-parameters of CT blocks by mimicking the GGO in the search space. At first, a population of solutions (Greylag geese) is randomly generated and every geese shows a different set of hyper-parameters like kernel size, number of attention heads, embedding dimensions, and learning rates. Here, the objective function considered is classification accuracy and it is expressed as:

$$Fitness = Maximize\ (Accuracy) \quad (6)$$

The GGO explored multiple candidate configurations across defined search spaces for identifying optimal settings that maximize validation accuracy. The following hyperparameters were considered: kernel size $\in$ {3, 5, 7}, number of attention heads $\in$ {4, 6, 8}, embedding dimension $\in$ {128, 256, 512}, and learning rate $\in$ {1e-3, 1e-4, 3e-4}. These ranges are chosen with respect to the combination of empirical trials and existing literature in transformer-based vision models. During each iteration, GGO evaluated candidate configurations using a fitness function and this automated approach eliminates manual tuning effort and improves model robustness.

GGO algorithm mimics the Greylag geese begins by randomly initializing a set of individuals. All individuals represent a potential solution to the given problem. The GGO population is $Y_j(j = 1, 2, 3, \ldots m)$ consists of m individuals. For assessing the quality of each solution, an objective function F_m is presented and shows each individual in the population. Once the F_m is computed for each agent Y_j, the header (best solution) is shown as Q. A key feature of the GGO is the Dynamic Groups behavior and it classifies the individuals into two sets: an exploration set (m_1) and exploitation set (m_2).

Exploration: The exploration is manageable for search space area location and prevents local optimized inactivity by moving to the ideal value.

Optimizing the search for the best solution: For enhancing exploration, the geese focus on finding promising new positions near the present position. This is achieved by continuously evaluating multiple nearby possibilities and choosing the most optimal one with respect to the value of the fitness. The GGO update the B and D by the $B = 2b \times r1 - b$ and $D = 2 \times r2$.

$$Y(l+1) = Y^*(l) - B|DY^*(l) - Y(l)| \tag{7}$$

where b decrease from 2 to 0; $r1$ and $r2$ random numbers between [0, 1].$Y^*(l)$ is the position of the leader,$Y(l)$ and $Y(l+1)$ are the agent at present iteration l and next iteration $l+1$.

For enhancing exploration and overcome search agents from being overly influenced by a single leader, three random agents (paddings) Y_{pad1}, Y_{pad2} and Y_{pad3} are selected. This process provides a more diverse search process. The current search agent's position is updated for $|B| \geq 1$.

$$Y(l+1) = u_1{}^*Y_{pad1} + x^*u_2{}^*\left(Y_{pad2} - Y_{pad3}\right) + (1-x)^*u_3{}^*\left(Y - Y_{pad1}\right) \tag{8}$$

The value of the x is given as:

$$x = 1 - \left(\frac{l}{l_{max}}()^2\right) \tag{9}$$

where l_{max} is the maximum iterations. The next updating process in which the b and B vector values are minimized for $r3 \geq 0.5$.

$$Y(l+1) = u_4{}^*|Y^*(l) - Y(l)| \times e^{at}cos(2\pi t) + [2u_1(r_4 + r_5)]^*Y^*(l) \tag{10}$$

where a is a constant term and the value of the t is [-1,1]. u_1, u_2, u_3 and u_4 are a parameter varies between [0,2] and r_4 and r_5 are the random numbers between [0, 1].

Exploitation: The exploitation phase focuses on enhancing present solutions for quality enhancement. In the GGO, the search agent with the high value of the fitness at the end of all iterations are identified and given priority. To achieve effective exploitation, GGO utilizes two processes; one is optimizing the search for the best solution and search the region over the best solution.

Optimizing the search for the best solution: The best solution is achieved by the below expression: The three sentries Y_{sen1}, Y_{sen2} and Y_{sen3} manage other individuals Y_{nonsen} for changing the positions to the defined prey position.

$$Y_1 = Y_{sen1} - B_1|D_1Y_{sen1} - Y| \tag{11}$$

$$Y_2 = Y_{sen2} - B_2|D_1Y_{sen2} - Y| \tag{12}$$

$$Y_3 = Y_{sen3} - B_3|D_1Y_{sen3} - Y| \tag{13}$$

where B_1, B_2, B_3 are computed as $B = 2b \times r1 - b$ and D_1, D_2, D_3 are computed as $D = 2 \times r2$. The position is updated by:

$$Y(l+1) = \overline{Y_j}\Big|_0^3 \tag{14}$$

Search the region over the best solution: **The most optimistic option is positioned near the header (optimal solution) during movement. This cause certain for exploring improvements by analyzing areas surrounding the best-known solution Y_{F1}. The position of the GGO is updated as:**

$$Y(l+1) = Y(l) + E(1+x)^*u^*(Y - Y_{F1}) \tag{15}$$

Choosing the best solution: **The GGO identifies the optimum solution by balancing exploration and exploitation. The GGO improves search diversity and overcomes the convergence issue by the mutation model and dynamic group adjustments. An elitist strategy shows that the header is retained and the position updated by:**

$$r1 = \frac{g}{1 - \frac{l}{l_{max}}} \tag{16}$$

where g is the constant value. For all iterations, the agents are repositioned, roles are interchanges and the best solution is obtained. Algorithm 1 shows the pseudocode of the GGO.

Algorithm 1: Pseudocode of the GGO

Initialize $Y_j(j = 1, 2, 3, \ldots m)$, m individuals and objective function F_m
Initialize $B_1, B_2, B_3, D_1, D_2, D_3, u_1, u_2, u_3$ and u_4
Compute F_m for each agent Y_j
Initialize Q
Update exploration set (m_1) and exploitation set (m_2)
While $l \leq l_{max}$
  for $(j = 1 : j < m_1 + 1)$
    when $(l\%2 == 0)$ **then**
      when $r3 \geq 0.5$ **then**
        when $|B| \geq 1$ **then**
          Update the position using $Y(l+1) = Y^*(l) - B|DY^*(l) - Y(l)|$
        else
          Choose Y_{pad1}, Y_{pad2} and Y_{pad3}
          Update $x = 1 - \left(\frac{l}{l_{max}}()^2\right)$
          Update the position using $Y(l+1) = u_1{}^*Y_{pad1} + x^*u_2{}^*\left(Y_{pad2} - Y_{pad3}\right) + (1-x)^*u_3{}^*\left(Y - Y_{pad1}\right)$
        end if
    else
        Update the position using $Y(l+1) = u_4{}^*|Y^*(l) - Y(l)| \times e^{at}cos(2\pi t) + [2u_1(r_4 + r_5)]^*Y^*(l)$
        end if
  end for
  for $(j = 1 : j < m_2 + 1)$ **do**
    when $(l\%2 == 0)$ **then**
      Compute $Y_1 = Y_{sen1} - B_1|D_1Y_{sen1} - Y|$, $Y_2 = Y_{sen2} - B_2|D_1Y_{sen2} - Y|$ and $Y_3 = Y_{sen3} - B_3|D_1Y_{sen3} - Y|$
      Update $Y(l+1) = \overline{Y_j}\Big|_0^3$
  else
    Update the position using $Y(l+1) = Y(l) + E(1+x)^*u^*(Y - Y_{F1})$
    end if
  end for
Compute F_m for each agent Y_j
Update the parameters
Set $l = l + 1$
When best F_m is similar as prior two iterations then
  Increase exploration set (m_1)
  Decrease exploitation set (m_2)
  end if
end while
Return Q

The CT-GGO model is designed with modular components where the CT architecture handles feature extraction and classification, and the GGO algorithm is an efficient optimizer of key hyperparameters such as the number of attention heads, kernel sizes, embedding dimensions, and learning rates. The CT component benefits from GGO by dynamically adapting these parameters for optimal learning, which enhances model robustness across different dataset configurations. When comparing other optimization and manual tuning, GGO introduces a bio-inspired, exploration–exploitation balance that accelerates convergence and reduces the risk of suboptimal performance. This integration is important to ensure that the hybrid architecture remains both efficient and accurate under varying training conditions.

4. Results and Discussion

The experimentation of the work is performed on Python platform. The experimentation is perform on 32 GB RAM, intel core i5 on Windows 10. The experimental hyper-parameters are given in Table 1.

For ensuring a fair and robust comparison, different DL models—CNN, EfficientNet-B0, MobileNetV2, Vision Transformer (ViT) and Convolutional Transformer (CT) are with consistent settings across the same training, validation, and test splits.

- The CNN model exploited in this work has five convolutional layers with ReLU activation functions. It is followed by max-pooling and fully connected layers with dropout regularization.
- EfficientNet-B0 and MobileNetV2 were initialized with ImageNet pre-trained weights, and their final layers were fine-tuned for the three-class KCN classification task. Both models used an input resolution of 224 × 224 pixels. The final layers are unfrozen and fine-tuned by the Adam optimizer with a learning rate of 1e-4.
- The ViT model (ViT-B/16 configuration) is also pre-trained on ImageNet-21 k and used 256 × 256 input images with a 16 × 16 patch size. It is fine-tuned with 12 transformer layers and 12 attention heads, using the AdamW optimizer with a learning rate of 3e-4. All models are trained for 100 epochs with a batch size of 32, and evaluated by different metrics for ensuring uniform performance assessment across architectures.
- CT model combines convolutional layers with attention mechanisms for capturing local and global features. It is trained under the same data conditions by the Adam optimizer with a learning rate of 3e-4. Unlike the proposed CT-GGO model, hyperparameters for the CT model are not optimized using GGO.

4.1. Datasets

The dataset comprised corneal images [Dataset] captured by various Pentacam devices configured with various settings like variations in the color scale steps utilized to visualize map. These color scales were with respect to the decimal grading system which displays measurements in microns for corneal elevation maps and width and in diopters for axial or sagittal curvature maps. An additional independent data obtained from a separate medical center in Brazil, was also employed for validation. In total, data from 204 eyes belonging to 104 individuals were categorized as normal, 215 eyes from 113 individuals were diagnosed with KCN, and 123 eyes from 63 individuals were classified as suspected case. The average ages (± standard deviation) for the normal, KCN, and suspect groups were 33.4 (± 10.1), 29.0 (± 9.3), and 28.6 (± 9.4) years. Further, 56 normal and 58 KCN eyes were imaged by a Pentacam device with various settings from the rest of the data. The independent validation data consisted of 150 eyes from 85 individuals, obtained at the de clinical center in Guarulhos, São Paulo, Brazil. This subset has 50 normal eyes from 29 participants, 50 KCN eyes from 31 patients, and 50 suspect case eyes from 25 patients. The average ages (± SD) in this dataset were 29.5 (± 4.7) for the normal group, 26.3 (± 6.8) for the KCN group, and 29.1 (± 5.3) for the suspect group. Fig. 3 shows the corneal maps from each class. (a) Normal cornea showing uniform curvature and thickness; (b) Suspect cornea with mild topographic irregularity; (c) KCN cornea with focal thinning, asymmetric curvature, and cone-like protrusion.

To ensure robust and unbiased model evaluation, the dataset is divided into training (70 %), validation (15 %), and test (15 %) subsets.

Table 1
Experimental hyper-parameters.

Hyper-Parameters	Values
Number of attention heads	4
Embedding dimension	256
dropout	0.1
Learning rate	3e-4
Optimizer	GGO
Batch size	32
Weight decay	0.01
Epochs	100
Population size	50
Iterations	100

4.2. Performance measures

Different performance measures like Accuracy, precision, Recall, specificity, False Negative Rate (FNR), Matthews Correlation Coefficient (MCC), confusion matrix, Receiver Operating Characteristic (ROC), accuracy-loss and t-SNE plot are measured. The expressions for computing these measures are expressed as:

Accuracy is the ratio of every accurately identified normal, KCN and suspect cases to the entire cases. It is given as:

$$Accuracy = \frac{T_x + T_y}{T_x + T_y + F_x + F_y} \tag{17}$$

Precision is rate of actual KCN cases among those identified as KCN by the approach. It is given as:

$$Precision = \frac{T_x}{T_x + F_x} \tag{18}$$

Recall is the potential of the approach to correctly find patients with KCN. It is given as:

$$R = \frac{T_x}{T_x + F_y} \tag{19}$$

Specificity is the potential of the approach to correctly find normal cases. It is given as:

$$Sp = \frac{T_y}{T_y + F_x} \tag{20}$$

FNR is the ratio of actual KCN cases that the approach incorrectly predicted as normal.

$$FNR = \frac{F_y}{T_x + F_y} \tag{21}$$

MCC is used for robustness of the model and it is expressed as:

$$MCC = \frac{(T_x T_y) - (F_x F_y)}{\sqrt{(T_x + F_x)(T_x + F_y)(T_y + F_x)(T_y + F_y)}} \tag{22}$$

The F-score is the harmonic mean of precision and recall; it is given as:

$$F - score = \frac{2T_x}{2T_x + F_x + F_y} \tag{23}$$

where T_x, T_y, F_x and F_y are the true positive, true negative, false positive and false negative.

4.3. Model interpretability

For understanding which regions of the corneal maps contribute most to the decisions of the model, Gradient-weighted Class Activation Mapping (Grad-CAM) is applied.

Fig. 4 shows the visualization of model interpretability using Grad-CAM for selected corneal topography images. Column (a) shows original input maps (e.g., anterior curvature), column (b) is the respective

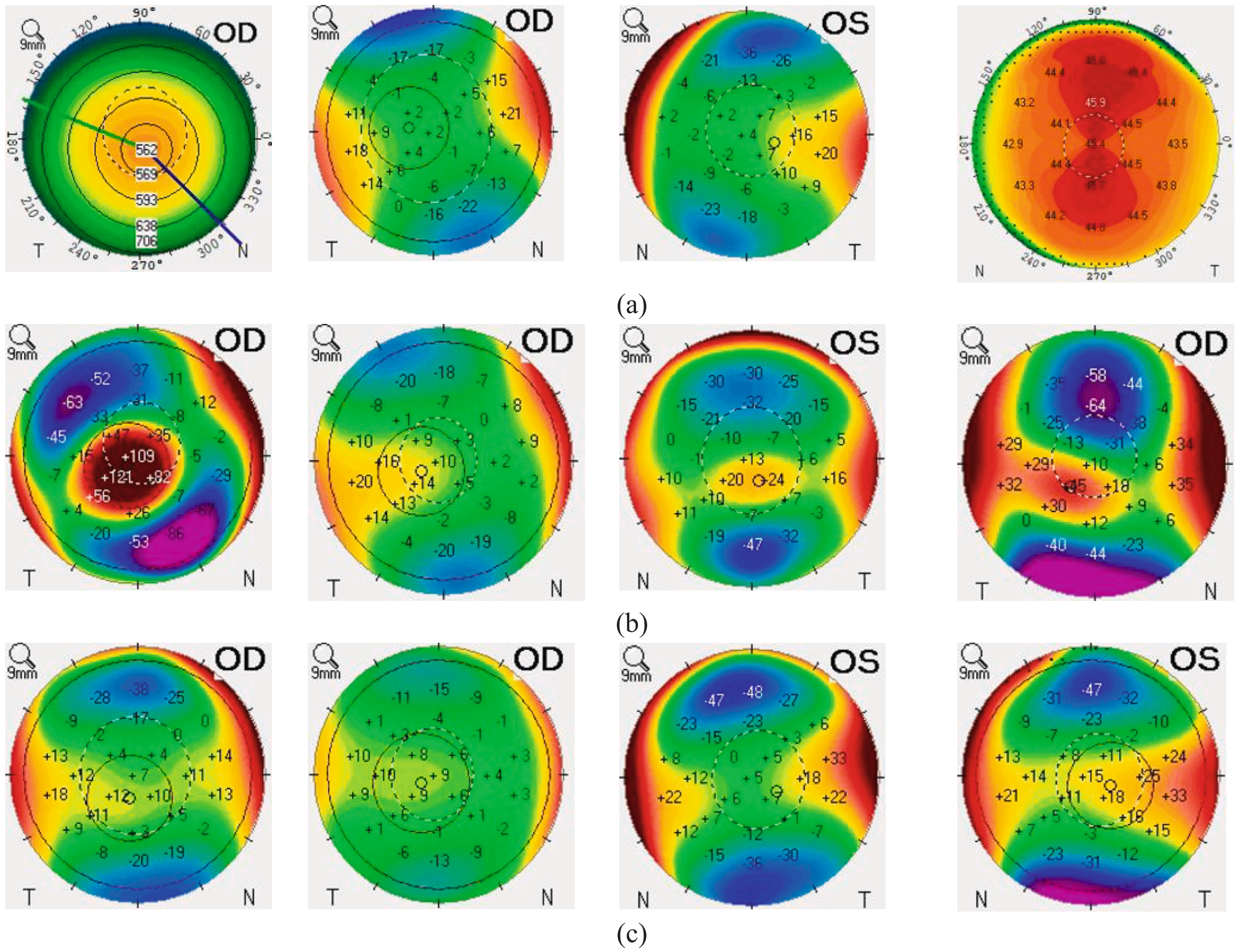

Fig. 3. Samples of the dataset (a) Normal, (b) KCN and (c) Suspect.

Grad-CAM activation heatmaps, and column (c) overlays the attention map on the original input. The red and yellow regions in column (b) represent areas where the CT-GGO model focuses during classification.

4.4. Ablation study

The following section shows the ablation study with the different DL models like CNN, efficientnet, MobileNet, ViT and the proposed CT-GGO by varying different training stages. Fig. 5 shows the accuracy comparison of the different DL models CNN, EfficientNet, MobileNet, ViT, and proposed CT-GGO over different training percentages (60 %, 70 %, 80 %, and 90 %). As the training percentage improves, all models exhibit advancement in accuracy. When, the training percentage is 90, the accuracy achieved by the CNN, EfficientNet, MobileNet, ViT, and proposed CT- GGO are 94.96 %, 95.12 %, 97.89 %, 98.4 and 99.21 %. It is observed that the CT-GGO model consistently outperformed the other DL models and attained high accuracy for all training level. The CNN shows the lowest accuracy (90 % to 94 %), EfficientNet attained accuracy (92 % to 95 %), MobileNet attained accuracy (93 % to 97 %), ViT attained accuracy (94 % to 98 %) and CT attained accuracy (94 % to 97 %). The outcomes show that higher training data leads to improved model performance and the suggested approach showed better effectiveness in KCN classification.

Fig. 6 delineates the recall comparison of the different DL models over various training percentages. All models exhibit an improvement in recall values when the training percentage increases. When, the training percentage is 60, the accuracy achieved by the CNN, EfficientNet, MobileNet, ViT, and proposed CT- GGO are 91.07 %, 93.94 %, 94.73 %, 94.96 % and 99.21 %. It is observed that the CT-GGO model consistently superior over all DL models. The CNN shows the lowest recall (91 % to 95 %), EfficientNet attained recall (93 % to 96 %), MobileNet attained recall (94 % to 97 %), ViT attained recall (94 % to 98 %) and CT attained accuracy (95 % to 97 %). This improvement recall values in CT- GGO is due to the model's integration of advanced feature extraction techniques and the optimization by GGO. However, other models obtained poor performance because of high complexity and over-fitting issues.

Fig. 7 delineates the precision comparison of the different DL models. The outcomes show that as the training percentage increases, precision enhance for each model. The CT-GGO consistently obtained high precision values of 95.63 % (training percentages is 60), 96.87 % (training percentages is 70), 97.85 % (training percentages is 80) and 98.96 % (training percentages is 90). The standard CNN is efficient, lacks in feature extraction and need extensive tuning and leads to less precision (91 % to 95 %). EfficientNet and MobileNet, designed for optimized performance and offered precision of (93 % to 96 %) and (94 % to 97 %). These models find difficulty in handling complex patterns. ViT shows strong performance (94 % to 98 %) because of the self-attention mechanisms however needs large data for effective training. The suggested model attained better outcomes because of the integration of the advantages of CT model with the GGO.

Fig. 8 delineates the specificity comparison of the different DL models. The specificity values attained by the CNN, EfficientNet,

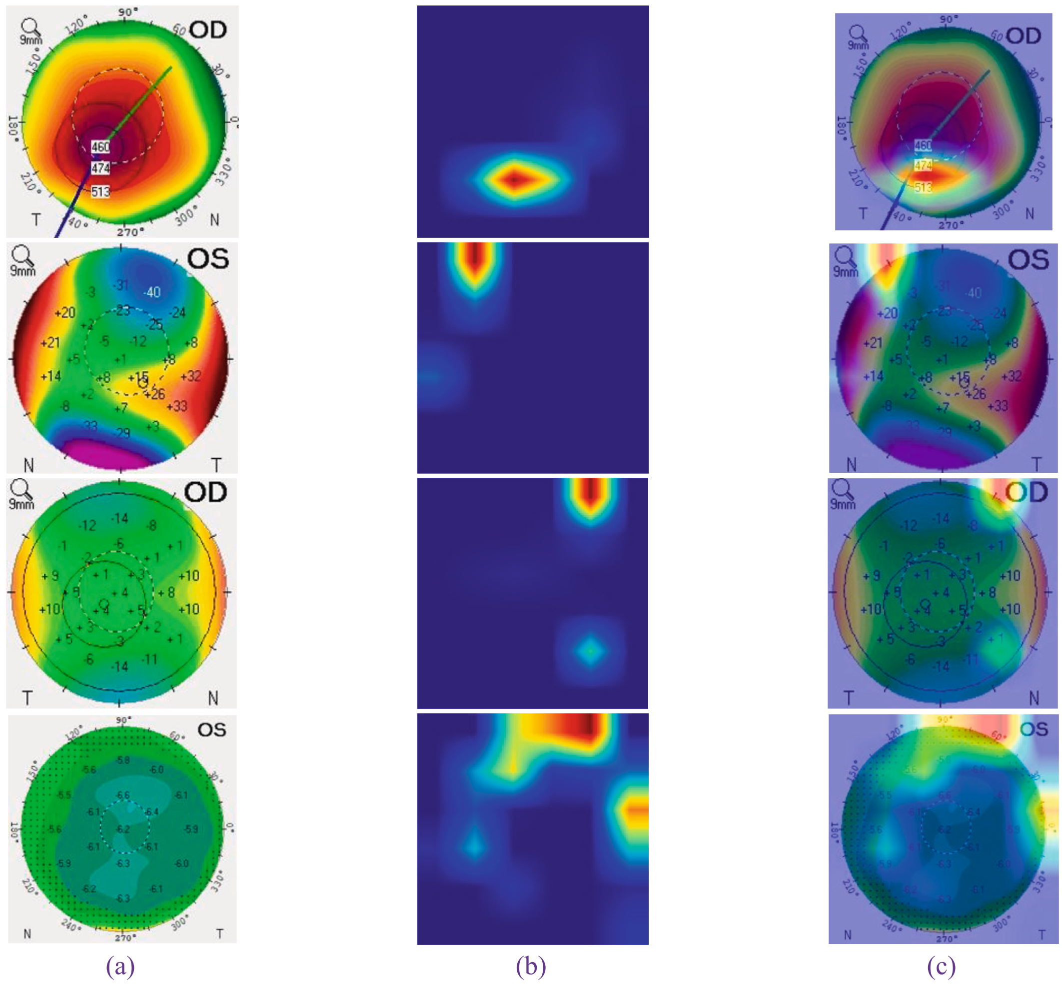

Fig. 4. Grad-CAM visualization for interpreting CT-GGO model predictions.

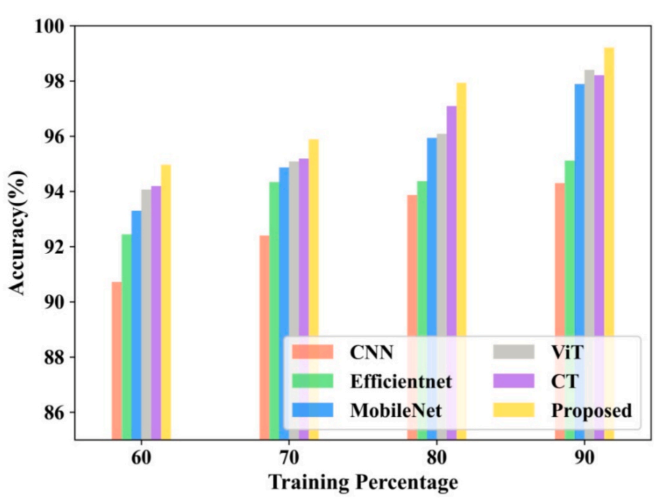

Fig. 5. Accuracy comparison.

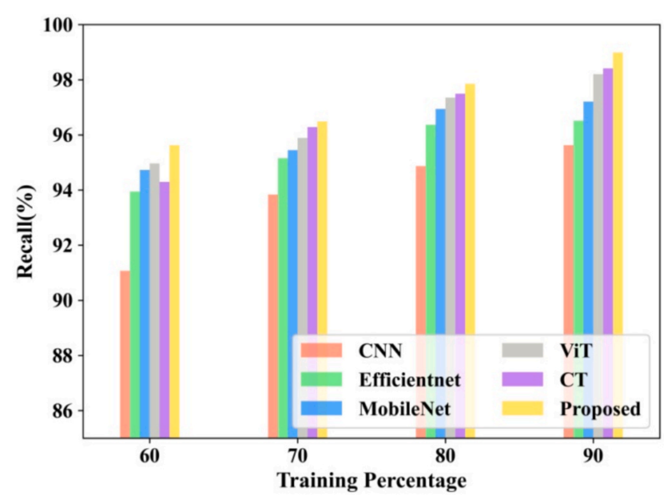

Fig. 6. Recall comparison.

MobileNet, ViT, CT and proposed CT-GGO are 90.95 %, 92.45 %, 93.3 %, 93.06 %, 93.6 % and 93.9 for training 60 %. The specificity values attained by the CNN, EfficientNet, MobileNet, ViT, CT and proposed CT-GGO are 92.4 %, 94.34 %, 94.87 %, 95.1 %, 95.3 % and 95.87 % for training 70 %. The specificity values attained by the CNN, EfficientNet, MobileNet, ViT, CT and proposed CT-GGO are 93.87 %, 94.37 %, 96.94 %, 97.2 %, 97.3 % and 97.98 % for training 80 %. Finally, when the training percentage is 90, the specificity values attained by the CNN, EfficientNet, MobileNet, ViT, CT and proposed CT-GGO are 94.3 %, 95.12 %, 97.98 %, 98.51 %, 98.6 % and 99 %. The outcomes show that the proposed CT-GGO is suitable for classifying KCN and assists ophthalmologists.

Fig. 9 delineates the FNR comparison of the different DL models over

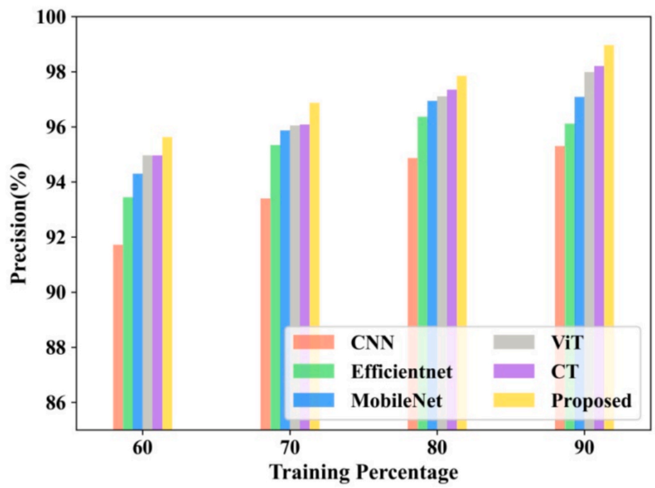

Fig. 7. Precision comparison.

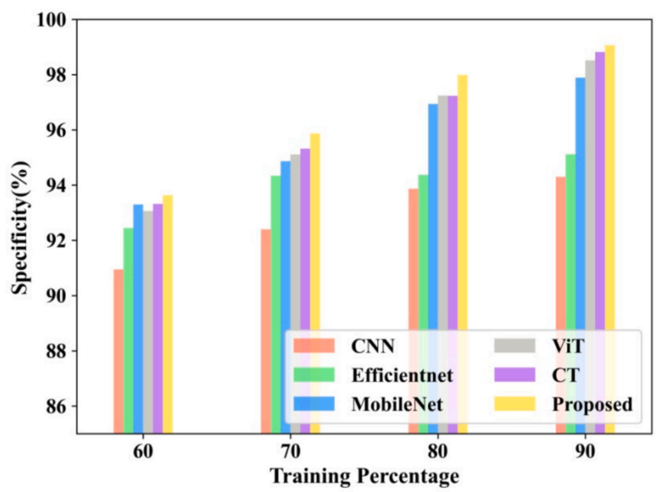

Fig. 8. Specificity comparison.

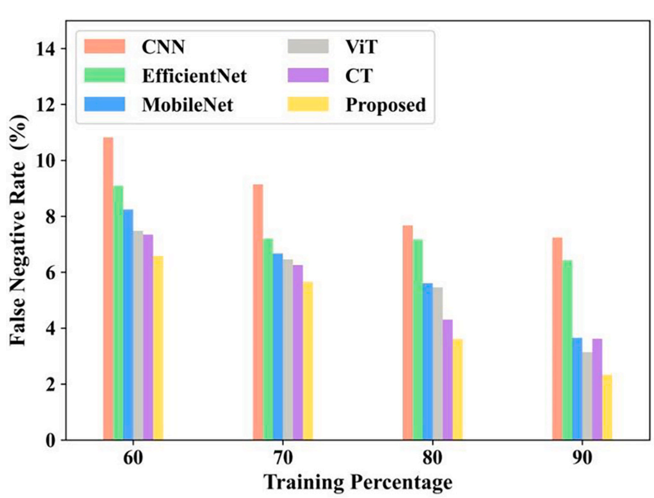

Fig. 9. FNR comparison.

different training percentages. All models show betterment in FNR values. When, the training percentage is 80, the FNR achieved by the CNN, EfficientNet, MobileNet, ViT, and proposed CT- GGO are 7.67, 7.17, 5.6, 5.4 and 3.6. It is observed that the CT-GGO model consistently superior over all DL models. The CNN shows the lowest FNR (7 % to 10 %), EfficientNet attained FNR (6 % to 9 %), MobileNet attained FNR (3 % to 8 %), ViT attained FNR (3 % to 7 %) and CT attained FNR (3 % to 7.5 %). The FNR values must be very low for the better disease classification model and it is observed that the suggested model attained better FNR values.

Fig. 10 compares the performance of different DL models by plotting the True Positive Rate (TPR) and False Positive Rate (FPR). The values of the Area Under the Curve (AUC) shows the entire classification performance and the high value of the AUC shows a better effective model. When comparing all approaches, CNN achieves an AUC of 0.979, ViT 0.973, MobileNet 0.972, EfficientNet 0.976 and CT 0.980 shows better but slightly varied performances. The proposed model outperformed all others with the highest AUC of 0.983 and showing superior discriminatory capability in differentiating positive and negative cases.

Fig. 11 shows the confusion matrix of the suggested CT-GGO model which classifies three classes like Normal, suspect and KCN. It is observed there are 480 samples are classified as normal and 15 samples are misclassified. Then, there are 290 samples are classified as suspect and 10 samples are misclassified. Finally, there are 250 samples are classified as KCN and 8 samples are misclassified.

Fig. 12 presents the training and validation accuracy and loss curves for the proposed CT-GGO model over 100 epochs. The accuracy graph in Fig. 12 (a) defines an increase in both training and validation accuracy, with the validation accuracy is superior over the training accuracy in beginning stage and provides better generalization. By the later epochs, both curves stabilize near to 100 % and shows effective learning. The loss graph in Fig. 12 (b) states a consistent decline in training and validation loss. The validation loss remains less than the training loss throughout and it show slight regularization effects or an optimally tuned model. No significant overfitting is observed, since both curves converge smoothly and confirms the robustness and efficiency of the CT-GGO in learning the data.

The Fig. 13 is a t-distributed Stochastic Neighbor Embedding (t-SNE) visualization, and is exploited to reduce dimensionality and visualize high-dimensional data in a 2D space. The plot reveals three well-separated, color-coded clusters with respect to the Normal (blue), Suspect (green), and KCN (red) classes. This clear separation shows that the model differentiates between these categories and demonstrates the strength of the learned feature representations. The discriminative clustering proves that the feature extraction and classification mechanisms in CT-GGO are functioning reliably and supporting its potential value in proper diagnosis of KCN.

Fig. 14 depicts the F-score comparison with respect to CNN,

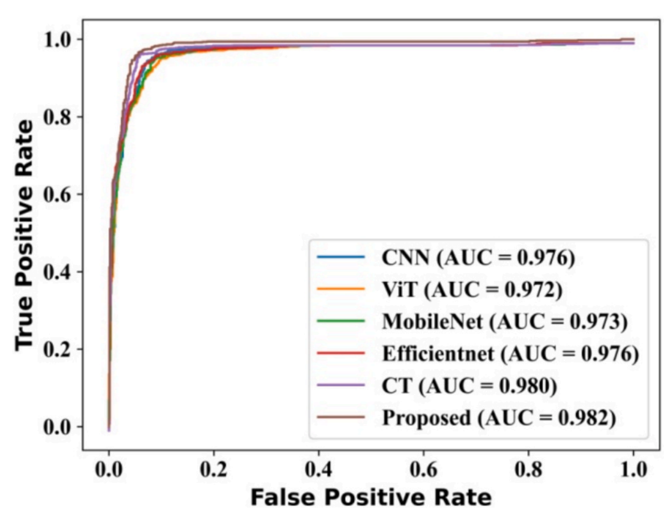

Fig. 10. ROC comparison.

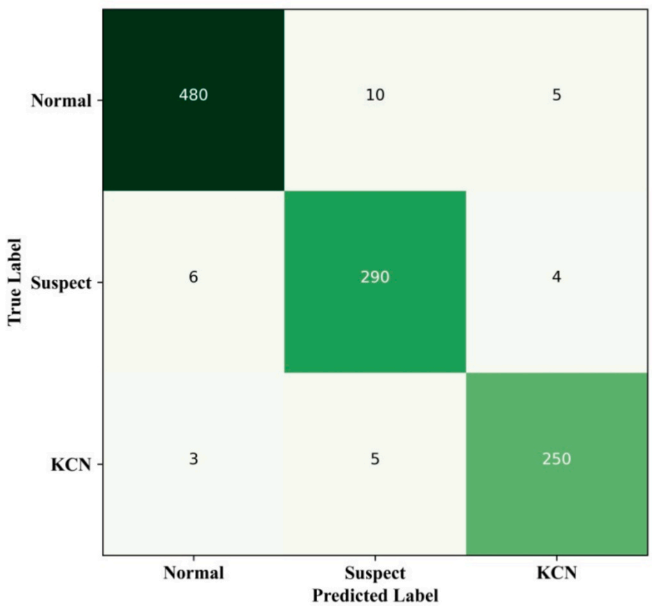

Fig. 11. Confusion matrix of the suggested CT-GGO.

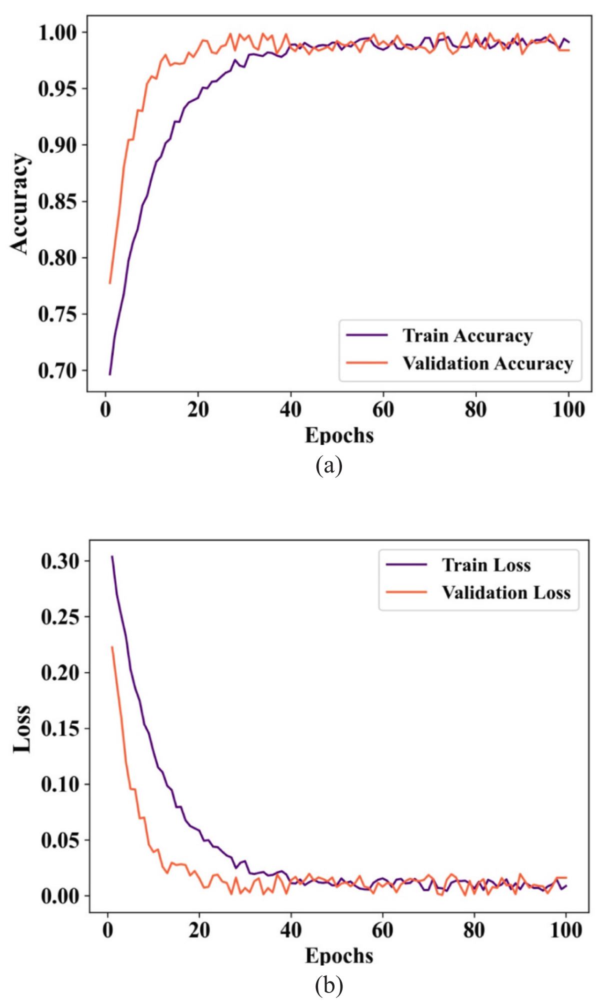

Fig. 12. Accuracy of the suggested CT-GGO.

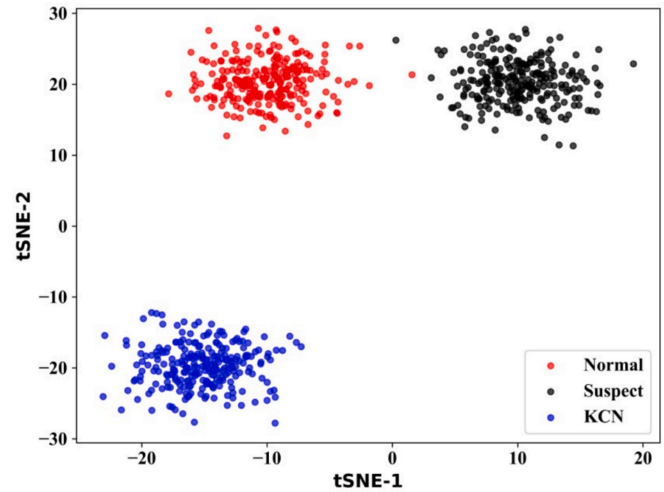

Fig. 13. t-SNE visualization.

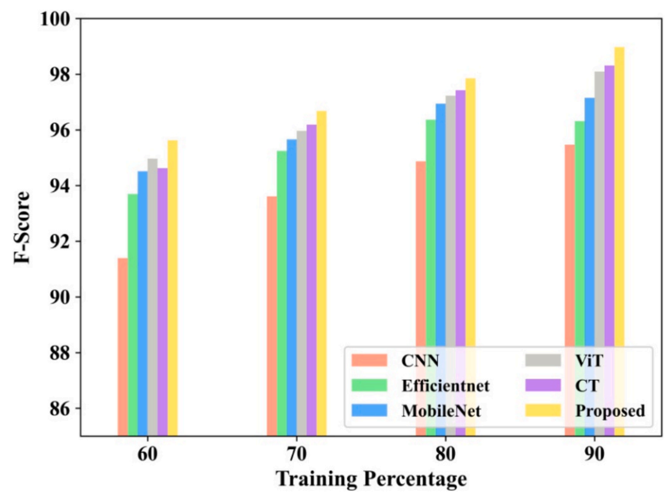

Fig. 14. F-score comparison.

EfficientNet, MobileNet, ViT, and the proposed CT-GGO on the basis of 60 %, 70 %, 80 %, and 90 % for KCN classification. The F-score consistently improves for all models and shows better classification accuracy. When the training percent is 80, the F-score value attained by the CNN is 94.87 %, EfficientNet is 96.37 %, MobileNet is 96.94 %, ViT is 97.22 %, ViT is 97.3 % and the proposed CT-GGO is 97.85 %. Similarly, when the training percent is 90, the F-score value attained by the CNN is 95.46 %, EfficientNet is 96.31 %, MobileNet is 97.194 %, ViT is 98.09 %, CT is 98.2 % and the proposed CT-GGO is 98.97 %.

Fig. 15 depicts the MCC comparison with respect to CNN, EfficientNet, MobileNet, ViT, and the proposed CT-GGO on the basis of training percentage. MCC must be high for better disease classification model. Across each training level, MCC improves as the amount of training data enhances and showing better classification reliability. MCC values attained by the proposed CT-GGO is 95.63 % (training is 60 %), 96.67 % (training is 70 %), 97.85 % (training is 80 %), and 98.97 % (training is 90 %),

Table 2 shows the comparison of model architectures in terms of parameters, FLOPs (Floating Point Operations per Second) and inference time. It is observed that the proposed CT-GGO attained parameter of 3.9 M, FLOPs of 0.65 GFLOPs and inference time of 0.08 ms. The proposed model aimed for enhancing efficiency by integrating an optimized architecture, reducing complexity, and attained less FLOPs with a faster inference time than other models. Even though the CT-GGO model has

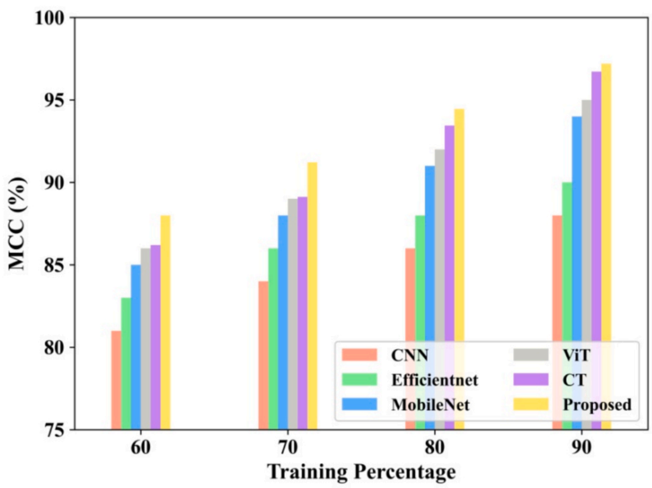

Fig. 15. MCC comparison.

Table 2
Comparison of model architectures.

Models	Parameters (M)	FLOPs (GFLOPs)	Inference time (ms)
CNN	20	5.5	0.27
ViT	86	17.6	0.50
MobileNet	4.2	0.57	0.10
EfficientNet	5.3	0.83	0.12
Proposed CT-GGO	**3.9**	**0.65**	**0.08**

slightly higher computational cost (0.65 GFLOPs) compared to MobileNet (0.57 GFLOPs), it provides better accuracy and diagnostic reliability. The faster inference time (0.08 ms) combined with higher diagnostic metrics justifies the computational tradeoff, particularly in clinical settings where decision accuracy and real-time screening are equally essential. In such scenarios, the minor increase in complexity is a worthwhile exchange for improved clinical outcomes.

Table 3 shows the outcomes of a Wilcoxon Signed-Rank Test comparing the $\rho -$ values of various DL approaches. The CNN and ViT models attained $\rho -$ values of 0.17 and 0.14and it indicates their statistical significance in performance comparison. Then, the MobileNet, EfficientNet, and a proposed model CT-GGO provided $\rho -$ values of 0.08, 0.06, and 0.04. This suggests that the CT-GGO has the most statistically significant difference compared to others. Lower $\rho -$ values shows better evidence over the null hypothesis and attained performance. These results show meaningful gains in diagnostic accuracy, particularly in recognizing borderline (suspect) cases. This statistical confirmation supports the clinical relevance of the suggested over existing methods.

Table 4 shows the comparison of the CT-GGO with recent works. It is noted that in this comparison also the suggested model attained better outcomes and suitable for KCN classification.

The results in Table 5 show that the CT-GGO model performs best when using the GGO optimizer, a learning rate of 3e-4, and 12 transformer layers and obtained the highest accuracy (99.20 %) and MCC (98.97 %). This confirms that both optimization strategy and model depth significantly influence classification performance and that the selected configuration offers optimal results. Table 6 depicts the performance of the proposed CT-GGO model across five cross-validation folds. The outcomes show consistent classification performance which shows robustness and generalizability across splits.

Table 3
Wilcoxon Signed-Rank Test.

Models	$\rho -$ value
CNN	0.17
ViT	0.14
MobileNet	0.08
EfficientNet	0.06
Proposed CT- GGO	0.04

Table 4
Comparison with recent works.

References	Accuracy	Recall	Specificity	AUC
[15]	0.94	0.94	0.97	0.98
[16]	95.31 %	98.71 %	–	–
[17]	–	–	–	95 %
[18]	92.13 %	92.49 %	91.54 %	0.971
[19]	–-	0.932	–	–
[20]	94.36	94.71 %	83.33 %	
[21]	0.89	–	–	0.94
[22]	93.1 %	86.6 %	98.2 %	–
[24]	98 %	–	–	0.93
Proposed	**99.2 %**	**98.9 %**	**99 %**	**0.983**

Table 5
Impact of different hyperparameter settings on CT-GGO model performance.

Optimizers	Accuracy (%)	Precision (%)	Recall (%)	F-score (%)	MCC (%)
Adam	98.87	97.45	98.90	97.80	97.10
Grid Search	97.62	96.10	97.90	96.80	95.90
SGD	96.42	95.10	96.12	95.50	94.20
Random Search	97.89	96.74	98.23	97.10	96.35
GGO	99.20	98.96	99.80	98.97	98.97
Learning Rate	Accuracy (%)	Precision (%)	Recall (%)	F-score (%)	MCC (%)
1e-3	97.31	96.12	97.40	96.50	95.40
1e-4	98.79	97.85	99.67	97.85	97.85
3e-4	99.20	98.96	99.80	98.97	98.97
Model Depth (Transformer layers)	Accuracy (%)	Precision (%)	Recall (%)	F-score (%)	MCC (%)
6	97.80	96.30	98.10	96.90	96.00
8	98.60	97.54	99.30	97.70	97.40
12	99.20	98.96	99.80	98.97	98.97

4.5. Discussion

Advantages and limitations of the proposed method.

The proposed CT-GGO model provides various advantages with respect to KCN classification. First, it achieves state-of-the-art accuracy, maintaining a lightweight architecture with only 3.9 million parameters and creating it more effective than heavy models like ViT and ensemble based models. The integration of convolutional layers with transformer blocks makes the network to obtain fine-grained spatial features and long-range dependencies. This is important in detecting complex structural deformations in corneal maps. Then, the use of GGO for hyperparameter tuning enhances robustness, and minimizes manual trial and error. But, the model does have some limitations. The transformer model

Table 6
Fivefold cross validation results for CT-GGO Model.

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC
Fold 1	98.90	98.80	98.70	98.75	0.991
Fold 2	99.00	98.90	99.10	99.00	0.993
Fold 3	99.10	98.95	99.20	99.07	0.994
Fold 4	99.30	99.10	99.40	99.25	0.996
Fold 5	99.00	98.85	99.00	98.92	0.992
Average	99.06	98.92	99.08	99.00	0.993

increase training time and computational resource demands, particularly in the optimization phase. Further, although GGO enhances parameter selection, it has additional complexity and may need proper control in limited resources. However, the trade-off is justified by the resulting boost in diagnostic accuracy and generalization. The CT-GGO model's "Suspect" classification is an important role in clinical triage workflows by determining ambiguous cases that may need expert review. This decision-support mechanism can help prioritize patient referrals and minimize the cognitive load on ophthalmologists. Furthermore, the low false positive rate of the model proves that unnecessary referrals are minimized, enhance overall efficiency and patient throughput in routine screening environments.

Implications for real-world clinical application

The CT-GGO model has strong potential for real-world deployment in ophthalmic screening systems, specifically for preoperative evaluation in refractive surgery. Its high classification accuracy, rapid inference time, and minimal parameter count make it suitable for real-time diagnostics, even in outpatient clinics or mobile diagnostic setups. The attention maps produced by transformer layers can provide interpretable insights into the decision-making process. This makes clinicians to analyze which regions of the corneal topography impact the classification process. Furthermore, the ability of the model to handle imbalanced datasets makes it applicable in cases where suspect cases are underrepresented like early screening programs. With further integration into electronic health record systems and diagnostic imaging devices, the CT-GGO model could consider as a valuable decision support tool, minimizing misdiagnosis rates and enhancing patient care outcomes.

5. Conclusion

This work presented a CT-GGO model for the classification of KCN disease. By integrating CNN with ViT for feature extraction and enhanced contextual learning, the proposed model attained better classification accuracy. The integration of the GGO optimizes hyperparameters effectively and enhances model performance and robustness. The accuracy, precision and AUC values attained by the suggested approach were 99.2 %, 98.9 % and 0.982 on the dataset from Kaggle. The findings show the potential of the CT-GGO for automated and early KCN detection and contribute to improved clinical decision-making and patient outcomes. In the future, the work will focus on expanding the dataset to combine a more diverse population for generalization enhancement. Further, combining multi-modal data like corneal topography, biomechanical properties, and ocular coherence tomography (OCT) scans, could enhance classification accuracy. Although, the strong performance of the proposed CT-GGO model, some challenges should be considered. The data, though comprehensive, may not fully show global clinical diversity, which could impacts the generalization of the model. Further, while attention mechanisms enhance model transparency to some extent, the interpretability of DL models remains a challenge in clinical settings. Overfitting is also a potential risk, particularly given the class imbalance and limited samples in some categories. These limitations will be overcome in future work through larger, multicenter data and by combining explainable AI frameworks for improving clinical reliability and trust. The demonstration of the CT-GGO model in real-time clinical settings will also be explored and ensures its applicability for ophthalmologists. As this work moves toward clinical application, future efforts will focus on aligning the CT-GGO system with regulatory models FDA and CE approvals, as well as IEC 62304 for medical software development. Moreover, rigid conformity to HIPAA and GDPR will be ensured for data privacy and ethical AI deployment. These steps are important for responsible integration into clinical environments.

CRediT authorship contribution statement

K. Balaji: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **N. Gobalakrishnan:** Writing – review & editing, Validation, Supervision, Resources, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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